

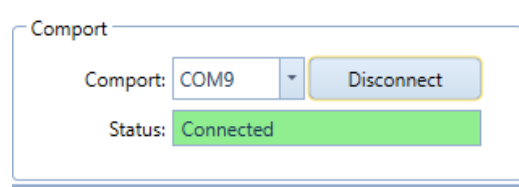
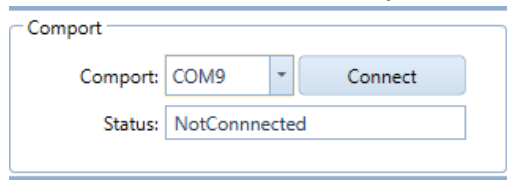
SOP for Active Place Avoidance (APA)

Zengyou Ye, Version 06/20/2025

1. Power On the APA Box →

2. Launch ConductAdapterUI Software

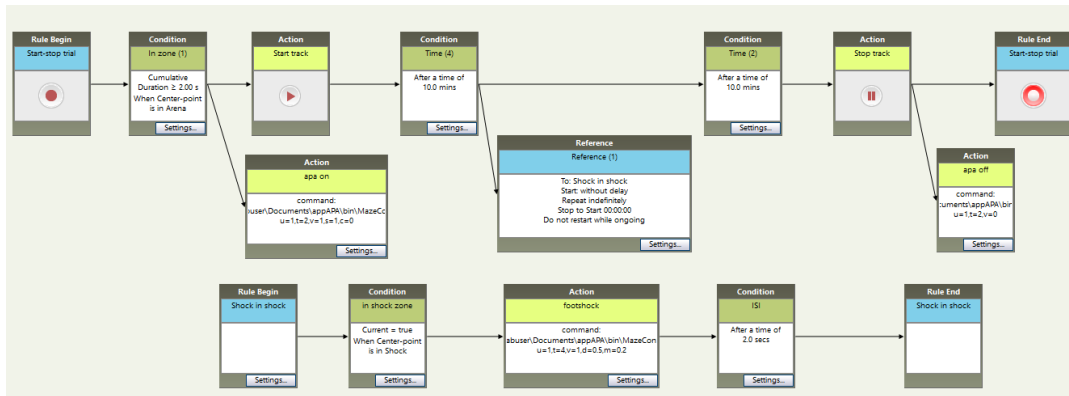
- On the computer, open the "ConductAdapterUI" application.
- In the interface, select **"COM9"** from the dropdown menu.
- Click **"Connect"** to establish communication with the APA system.
- Minimize the "ConductAdapterUI" window after the status turns **"Connected"**



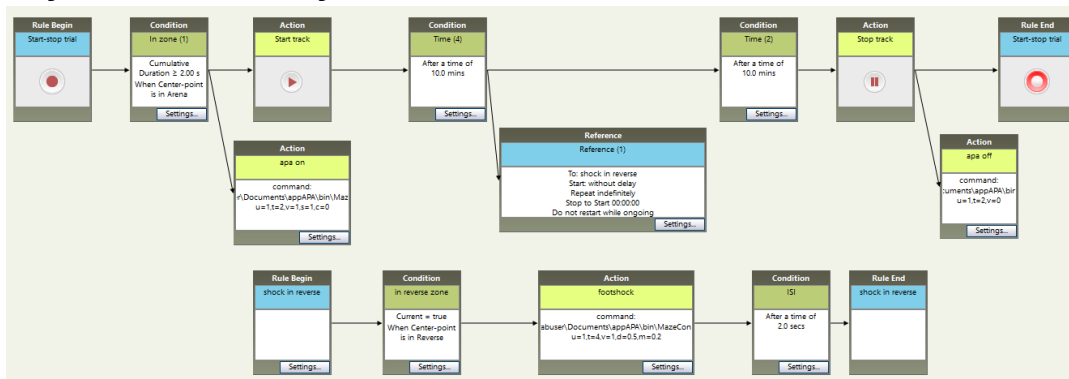
3. Open EthoVision XT with the Appropriate Protocol

Recommend: create a new copy of the protocol file for each day to keep records organized.

• Day 1: 10min Habituation and 10min Shock Training



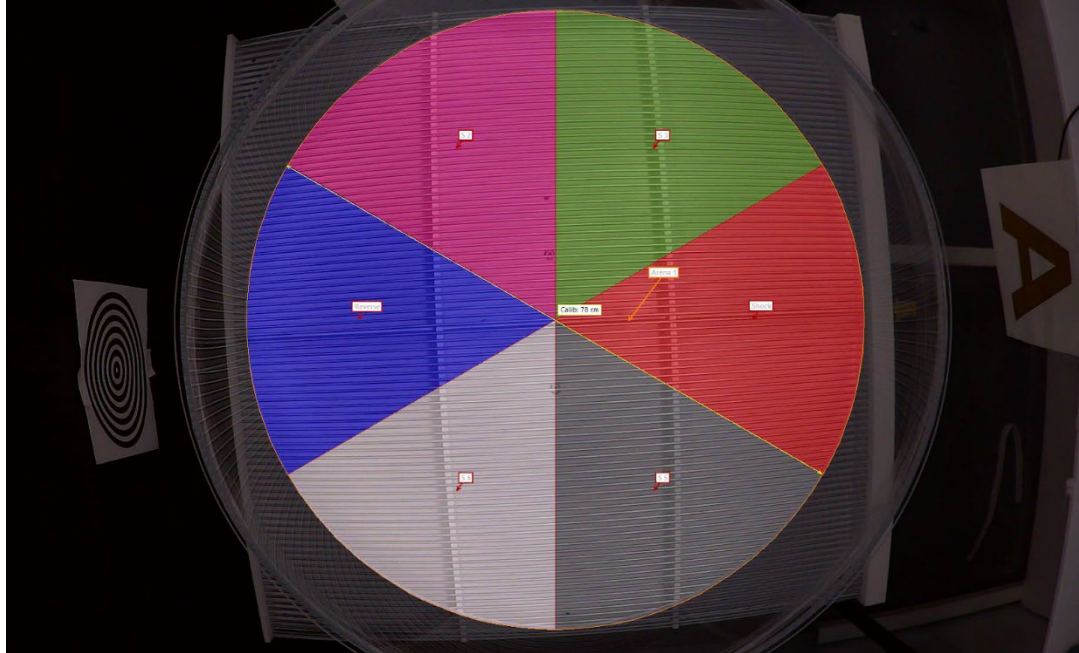
• Day 2: 10min Memory Recall and 10min Reversal Test



Optional: For extended memory testing, you may repeat the Day 1 protocol over multiple days.

4. Preparation Before Running

- Double-check the **camera view** and ensure it is properly aligned.



- Enter the **Mouse ID** into the software.
- Click **Run** to start the program.

5. Mouse Placement

Gently place the mouse into the APA chamber, positioning it on the **opposite side of the shock-delivery zone** and **facing the wall**.

6. Start of Trial

The chamber will begin rotating automatically once **EthoVision** detects the mouse.

Note:

Type	Action	Command	Note	Example
Shock	Start		u=1,...,4 t=4 type shock v=1 start d=1-60 for duration in seconds m=0.1 - 4.0 for current level in mA, step 0.1	u=1,t=4,v=1,d=5,m=0.4
	Stop	u=1,t=4,v=0	u=1,...,4 t=4 type shock v=0 stop	

Rotate	Start		u=1,...,4 t=2 type rotate v=1 start s=4-100 speed in round/minute c=0 or 1 for direction, 0 counterclockwise and 1 clockwise	u=1,t=2,v=1,s=20,c=0
	Stop	u=1,t=2,v=0	u=1,...,4 t=2 type rotate v=0 stop	